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SHORT COMMUNICATION

RRFT1 (Redox Responsive Transcription Factor 1) is involved in extracellular ATP-regulated gene expression in *Arabidopsis thaliana* seedlings

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ABSTRACT

As an apoplast signal molecule, extracellular ATP (eATP) is involved in the growth regulation of Arabidopsis thaliana seedlings. Recently, RRFT1 was revealed to be involved in eATP- regulated seedling growth. To further verify the role of RRTF1 in seedlings' eATP response, expression of 20 eATP-responsive genes in wild type (Col-0) and RRTF1 null mutant (rrtf1-1) seedlings were investigated by using realtime quantitative PCR. After 0.5 mM ATP stimulation, the response of these genes' expression in rrtf1-1 seedlings was significantly different from that in Col-0 seedlings. Proteins which are encoded by these genes include transcription factors, plasma membrane receptors like kinases, ion influx/efflux transporters and hormone signaling components. The results indicated that RRTF1 may be involved in eATP regulated physiological responses via regulating the expression of some functional genes.

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As an apoplast signal molecule, extracellular ATP (eATP) is involved in the growth regulation of *Arabidopsis thaliana* seedlings. Recently, RRFT1 was revealed to be involved in eATP- regulated seedling growth. To further verify the role of RRTF1 in seedlings' eATP response, expression of 20 eATP-responsive genes in wild type (Col-0) and *RRTF1* null mutant (*rrtf1-1*) seedlings were investigated by using realtime quantitative PCR. After 0.5 mM ATP stimulation, the response of these genes' expression in *rrtf1-1* seedlings was significantly different from that in Col-0 seedlings. Proteins which are encoded by these genes include transcription factors, plasma membrane receptor-like kinases, ion influx/efflux transporters, and hormone signaling components. The results indicated that RRTF1 may be involved in eATP regulated physiological responses via regulating the expression of some functional genes.

Extracellulr ATP, which is secreted from the cytoplasm, acts as a signaling molecule in the extracellular matrix. eATP is involved in regulating vegetative growth, reproductive processes, and stress responses.[1–4] Functional gene expression and protein synthesis may be a central event in these eATP-stimulated responses.[5,6] Hence, transcription factors may play a key role in eATP-regulated physiological processes as well. Some ethylene-responsive factors were reported to be induced by eATP and involved in eATP signaling.[7–9] Recently, RRTF1, an ethylene-responsive transcription factor, was revealed to participate in eATP-regulated root & hypocotyl growth via regulating auxin transport and functional genes' expression.[10] In 2017, in an DNA microarray assay, we found some ATP responsive genes which may possibly be involved in eATP-regulated root growth.[9] To further clarify the role of RRTF1 in eATP signaling, expression of these genes in eATP-treated wild type (Col-0) and RRTF1 null mutant (*rrtf1-1*) seedlings were detected by using realtime quantitative PCR.

Arabidopsis thaliana seeds were surface-sterilized and sown on the surface of solid 1/2 MS medium (containing 0.8% phytagel) in square culture dishes. After 2 days vernalization at 4°C, culture dishes were vertically placed in a growth chamber which was running at 22°C, 130 μmol/m^2·s illumination and 16/8 h light/dark ratio. 4-d-old seedlings were then transplanted onto 0.5 mM ATP-containing medium which was made according to Zhu et al.[10] 1/2 MS medium was pre-adjusted to maintain the final pH value around 5.8. 2 or 3 h after transplantation, seedlings were harvested and rapidly frozen in liquid nitrogen for RNA extraction. Total RNA was isolated from 0.1 g root tissue using the RNAiso plus reagent, and 2 μg of total RNA was reverse-transcribed to cDNA using M-MLV reverse transcriptase. Realtime quantitative PCR was performed using an ABI7500 system, *ACTIN2* was used as the constitutive reference gene. The primers' sequences are listed in Table S1.

The results showed that, after ATP stimulation, 20 genes' expression pattern were markedly different in Col-0 and *rrtf1-1* seedlings. The responsiveness of these genes' expression can be sorted into three categories: (1) 14 genes' expression were remarkably up-regulated in Col-0 while not up-regulated or even down-regulated in *rrtf1-1*, e.g. *ABCG39, ABCG40, XTH33, GH3.4*, etc.; (2) 3 genes' expression were not or only slightly up-regulated in Col-0 while remarkably down-regulated in *rrtf1-1*, e.g. *LECRK13*, *XTH22*, and *CRK5*; (3) 3 genes' expression were remarkably down-regulated or not changed in Col-0 while markedly up-regulated in *rrtf1-1*,

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e.g., *ACA13*, *SAUR68*, and At3g28580 (Figure 1). The results indicated that RRTF1 may be involved in eATP regulated gene expression as a positive or negative regulator.

Based on the data from the NCBI (National Center for Biotechnology Information) and UniProt (Universal Protein) website, proteins which are encoded by these genes can be categorized into five groups: (1) Transcription factors, 3 ethylene-responsive transcription factors (ERF018, ERF105, and ERF114) are included; (2) Cell wall construction involved enzymes, XTH22, and XTH33 are included; (3) Plasma membrane lectin-domain containing receptor kinases, LECRK13, and LECRK59 are included; (4) Plasma membrane transporters, 4 ABC (ATP Binding Cassette) transporters (ABCA8, ABCG11, ABCG39, and ABCG40), an organic cation transporter (OCT1) and a Ca^{2+} ATPase (ACA13) are included; (5) Hormone metabolism and signaling related proteins, an IAA-amido synthase (GH3.4), 3 auxin-responsive proteins (SAUR68, SAUR76, and ARGOS), 2 ABA signaling related proteins (PRN1 and At3G28580) and 1 cytokinin responsive factor (CRK5) are included (Table 1).

These proteins are involved in multiple physiological processes, including cell division & differentiation, and stress (or defense) responses. eATP-regulated expression of these genes may be involved in eATP-regulated seedling growth; hence, the different responses of Col-0 and *rrtf1-1* seedlings to eATP stimulation[10] may partially result from the differential expression

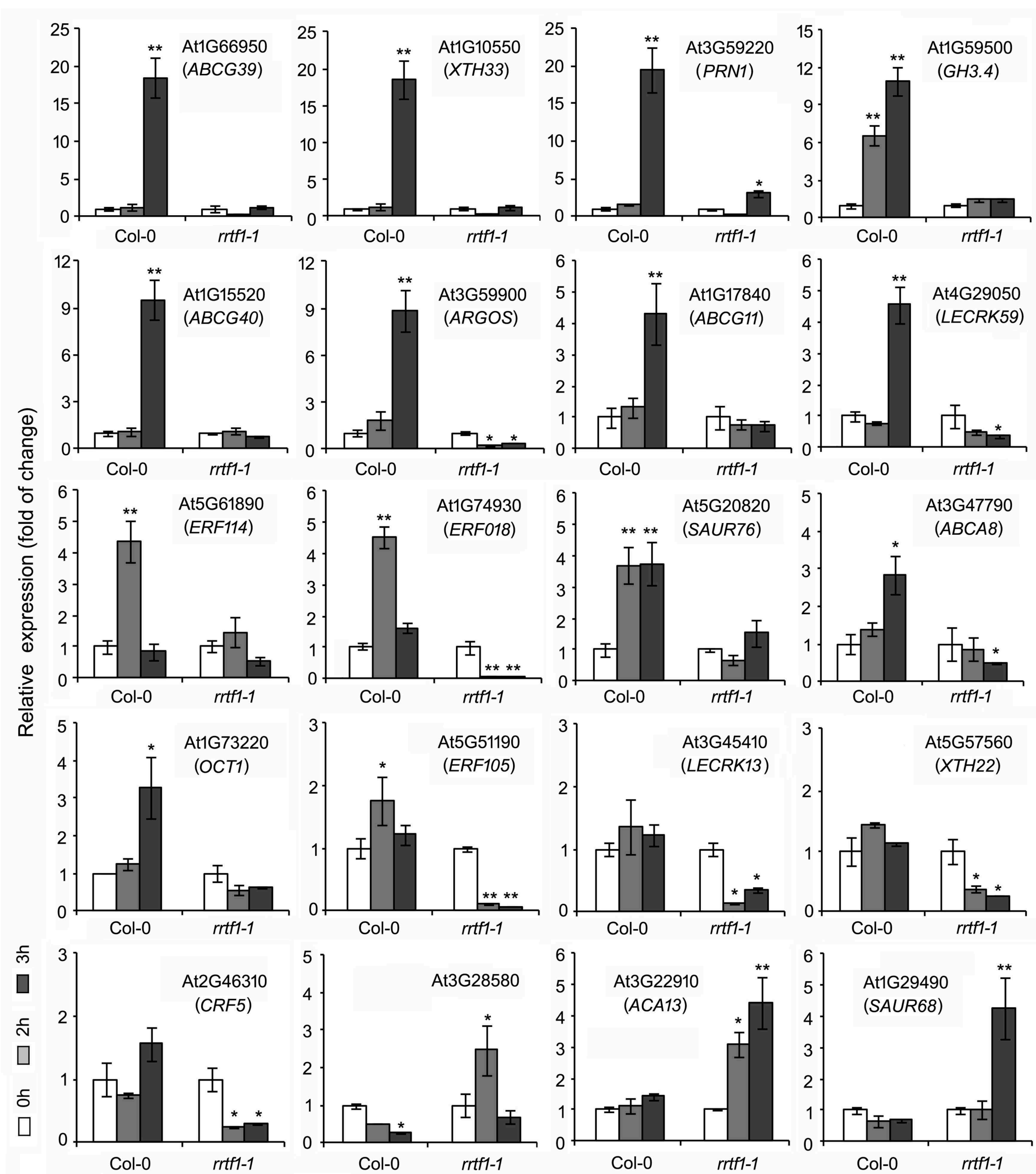

Figure 1. The expression of 20 eATP responsive genes in Col-0 and *rrtf1-1* seedlings. Expression of gene was detected by using realtime quantitative PCR, which *Actin2* was used as a reference gene. The primers of these genes are listed in Table 2. Four-d seedlings were transplanted onto 0.5 mM ATP-containing medium and cultured under light for 2 or 3 h. Then, total RNA was extracted and expression of each gene was detected. Data from three biological replicates (each include three technical replicates) were used to calculate the mean ± SD and statistically analyzed. Student's *t*-test *p*-values: * $p < .05$, ** $p < .01$.

Table 1. Function analysis of RRTF1-regulated eATP-responsive genes. Description of the genes' function is based on information from website of NCBI (The National Center for Biotechnology Information) and UniProt (Universal Protein).

Gene Locus	Name	Summary for Possible Function
Transcription Factors		
At5G51190	*ERF105*	ethylene responsive transcription factor, involved in ethylene signaling
At5G61890	*ERF114*	ethylene responsive transcription factor, involved in ethylene signaling
At1G74930	*ERF018*	a member of DREB transcription factor family, involved in ethylene signaling
Cell Wall Construction		
At1G10550	*XTH33*	xyloglucan:xyloglucosyl transferase 33, involved in cell wall modification
At5G57560	*XTH22*	xyloglucan endotransglucosylase/hydrolase 22, involved in cell wall construction
Plasma Membrane Receptors		
At4G29050	*LECRK59*	L-type lectin-domain containing receptor serine/threonine kinase V.9
At3G45410	*LECRK13*	LECRK I.3, involved in resistance to pathogenic fungus
Cell Membrane Transporters		
At1G66950	*ABCG39*	involved in ABA transport and resistance to lead (Pb^{2+})
At1G15520	*ABCG40*	a pump to exclude Pb^{2+} and/or Pb^{2+}-containing compounds from the cytoplasm
At3G47790	*ABCA8*	ABC2 homolog 7, ATPase-coupled transmembrane transporter
At1G17840	*ABCG11*	ABC transporter in charge of cutin transport to extracellular matrix
At1G73220	*OCT1*	Organic Cation Transporter 1, involved in lateral root development
At3G22910	*ACA13*	a putative plasma membrane-type calcium-transporting ATPase
Plant hormone metabolism or signaling		
At3G59900	*ARGOS*	involved in auxin-regulated cell proliferation, lateral organ growth
At1G59500	*GH3.4*	an IAA-amido synthase, conjugates amino acids to IAA to inhibit its activity
At1G29490	*SAUR68*	SAUR-like auxin-responsive protein, promote auxin-stimulated organ elongation
At5G20820	*SAUR76*	SAUR-like auxin-responsive protein, positive regulator of cell growth
At3G59220	*PRN1*	Pirin-1, involved in ABA induced responses
At2G46310	*CRK5*	cytokinin responsive factor, involved in cotyledon, leaf & embryo development
At3G28580	-	an AAA-type ATPase, involved in response to ABA

pattern of these genes in seedlings. Recently, by using chromatin immunoprecipitation (CHIP) and high throughput sequencing analysis, 48 genes were screened out to be ATP responsive genes which may be directly regulated by RRTF1.[10] The results here provided additional evidence that RRTF1 may be involved in eATP signaling. Nevertheless, the 20 genes we detected here were not enriched in CHIP assay, so RRTF1 may possibly regulate the expression of these genes indirectly.

During the past 2 decades, it was revealed that seedlings responded to the high concentration of eATP as a suppressed growth rate and changed the growth direction of roots or hypocotyls.[7–12] Plant hormone (auxin, jasmonic acid, ethylene) signaling components were revealed to be involved in eATP regulated seedling growth.[6–14] The 20 proteins had been proved or predicted to be involved in stimuli-regulated vegetative organs growth. Results here provided further evidence that these proteins may play roles in eATP signaling as well. RRTF1 has been revealed to be involved in plant growth and stress responses.[15,16] Results here provided new evidence to elucidate the role of RRTF1 in Arabidopsis seedlings' response to eATP. It is of good help for understanding the role of ethylene-responsive transcription factors in the signaling of apoplast messengers. Recently, it was revealed that MYC and CAMTA3 transcription factors play essential roles in regulating eATP responsive genes' expression.[6] It is expected that the discovery of more transcription factors in eATP signaling will provide valuable information to elucidate the mechanism of eATP- regulated gene expression.

Disclosure of potential conflicts of interest

No potential conflicts of interest were disclosed.

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